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MASTER EXECUTIVE PORTFOLIO & FINANCIAL ROADMAP

Popular Auto Spare ERP — Complete 9-Module Master Milestone & Budget Summary

Executive consolidated milestone breakdown, multi-module timeline mapping, engineering resource matrix, and grand commercial investment schedule for **Popular Auto Spare & A/C Parts Co. W.L.L.** Prepared by **SaNDs Lab Middle East W.L.L** in strategic alignment with **UniGlobal Consultancy**.

 TOTAL MODULES

9 Modules

End-to-End Automotive ERP

 TOTAL INVESTMENT

BD 34,090.910

Fixed Baseline in Bahraini Dinars

 CUMULATIVE EFFORT

88 Weeks

17.6 Engineering Months

 DEDICATED TEAM

5 Specialists

100% Dedicated Full-Time

VERIFICATION TRANCHES

35 Gates

Zero-Risk Milestones



9/9

Modules Documented

Complete functional & technical specs



BD 1,363.636

Monthly Team Burn-Rate

Fixed transparent rate card



100% Multi-Branch

Enterprise Coverage

Warehouses, POS counters, mobile floor



15-Day UAT

Acceptance Grace SLA

Rigorous milestone verification

Total Fixed Project Investment (All 9 Modules)

Unified contract value covering the complete design, engineering, multi-branch testing, data migration, hardware integrations, training, and production deployment across all 9 modules for Popular Auto Spare & A/C Parts Co. W.L.L.

BD 34,090.910

THIRTY-FOUR THOUSAND NINETY BAHRAINI DINARS & 910 FILS

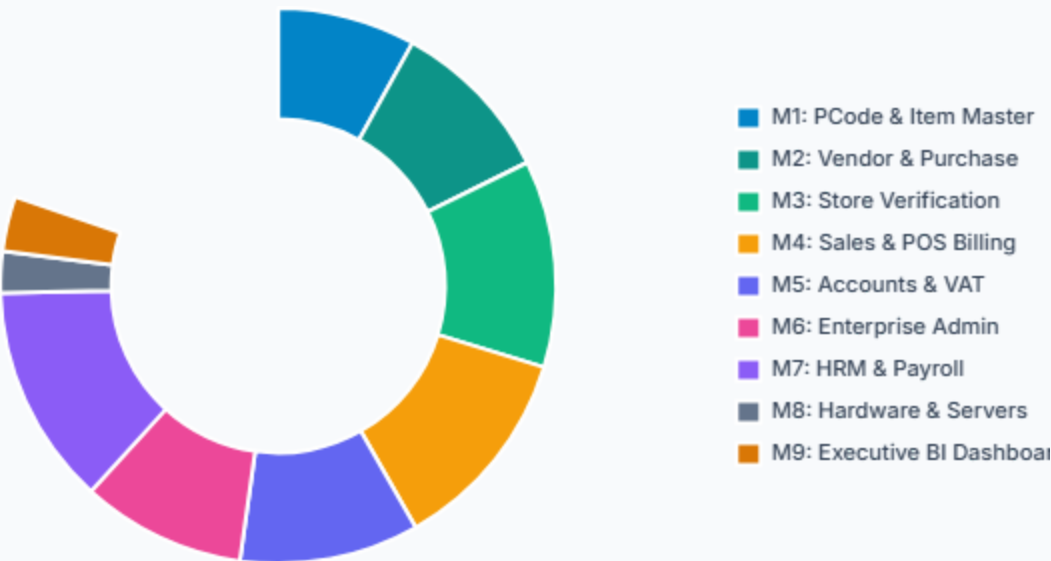
Visual Milestone & Budget Analytics

Visualizing budget distribution and engineering effort allocation across all 9 modules

INTERACTIVE VISUAL
INSIGHTS

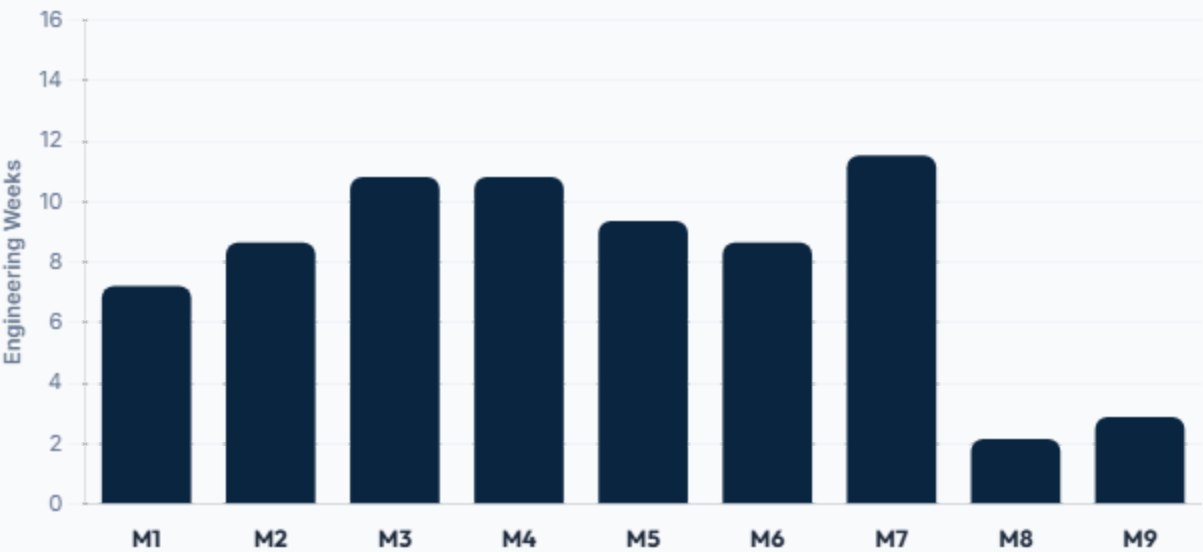
 Budget Distribution by Module (BHD)

Total: BD 34,090.910



 Engineering Timeline by Module (Weeks)

Total: 88 Working Weeks





Master 9-Module Budgeting & Milestone Matrix (BHD)

ALL AMOUNTS IN BAHRAINI DINARS (BHD)

At-a-glance comparative overview of all 9 modules, timelines, effort, deliverables, and total fees in Bahraini Dinars

#	MODULE SCOPE & IMPLEMENTATION DOMAIN	TIMELINE	EFFORT	MILESTONE GATES	TOTAL COST (BHD)	WEB BUR
1	SL-POP-ERP-MS-001 BA Ref: DOC-001 v1.0 PCode Generation, Cataloguing & Item Master Engine	10 Weeks	2.50 Mo	4 Gates (25%)	BD 3,409.091	BD 340
2	SL-POP-ERP-MS-002 BA Ref: DOC-002 v1.0 Vendor & Purchase Management, Supplier Portal & 3-Way Match	12 Weeks	3.00 Mo	4 Gates (25%)	BD 4,090.909	BD 340
3	SL-POP-ERP-MS-003 BA Ref: DOC-003 v1.0 Store Verification, Multi-Warehouse Stock & Location Matrix	15 Weeks	3.75 Mo	4 Gates (25%)	BD 5,113.636	BD 340
GRAND TOTAL: COMPLETE 9-MODULE AUTOMOTIVE ERP PORTFOLIO		88 Weeks	17.60 Mo	35 Gates	BD 34,090.910	BD 387

#	MODULE SCOPE & IMPLEMENTATION DOMAIN	TIMELINE	EFFORT	MILESTONE GATES	TOTAL COST (BHD)	WEEKLY BURDEN
4	SL-POP-ERP-MS-004 BA Ref: DOC-004 v1.0 Sales Process, Mobile POS, Multi-Branch Billing & Return Control	15 Weeks	3.75 Mo	4 Gates (25%)	BD 5,113.636	BD 340
5	SL-POP-ERP-MS-005 BA Ref: DOC-005 v1.0 Accounting & Financial Management, General Ledger, AP/AR & VAT	13 Weeks	3.25 Mo	4 Gates (25%)	BD 4,431.818	BD 340
6	SL-POP-ERP-MS-006 BA Ref: DOC-006 v1.0 Enterprise Administration, Facility, Fixed Assets, Fleet & Vault	12 Weeks	3.00 Mo	4 Gates (25%)	BD 4,090.909	BD 340
7	SL-POP-ERP-MS-007 BA Ref: DOC-007 v1.0 Human Resource Management, Biometrics, Leave, Payroll & Gratuity	16 Weeks	4.00 Mo	4 Gates (25%)	BD 5,454.548	BD 340
8	SL-POP-ERP-MS-008 BA Ref: ARCH-001 v1.0 Hardware Setup, Handheld QR	3 Weeks	0.75 Mo	3 Gates (33%)	BD 1,022.727	BD 340
GRAND TOTAL: COMPLETE 9-MODULE AUTOMOTIVE ERP PORTFOLIO		88 Weeks	17.60 Mo	35 Gates	BD 34,090.910	BD 387

#	MODULE SCOPE & IMPLEMENTATION DOMAIN	TIMELINE	EFFORT	MILESTONE GATES	TOTAL COST (BHD)	WEEK BUR
Devices, ESC/POS & Cloud Cluster						
SL-POP-ERP-MS-009						
BA Ref: DOC-009 v1.0						
9	Executive Management Dashboard, 8-Module BI & Mobile Cockpit	4 Weeks	1.00 Mo	4 Gates (25%)	BD 1,363.636	BD 340
GRAND TOTAL:						
COMPLETE 9-MODULE AUTOMOTIVE ERP PORTFOLIO		88 Weeks	17.60 Mo	35 Gates	BD 34,090.910	BD 387



9-Module Architectural Scope & Deliverable Pillars

FUNCTIONAL PORTFOLIO

Key functional scope, core modules, and business impact for Popular Auto Spare Co.

MODULE 1

BD 3,409.091

PCode Generation & Universal Item Master

Automated smart PCode generation algorithms, multi-brand hierarchy (Car Make/Model/Year), 1-to-Many OEM cross-reference matrix, 50,000+ SKU legacy data cleansing, and sub-100ms multi-parameter fuzzy search.

🕒 10 Weeks (2.5 Mo)

MODULE 2

BD 4,090.909

Vendor & Purchase Management with 3-Way Match

Multi-country vendor master, tokenized supplier RFQ portal, MOQ reorder buffering, CTO purchase approvals, automated PO splits by brand, Goods Receipt Note (GRN) with landed cost, and 3-way invoice matching.

🕒 12 Weeks (3.0 Mo)

MODULE 3

BD 5,113.636

Store Verification & Multi-Warehouse Stock Control

5-tier warehouse spatial layout (Zone/Aisle/Rack/Shelf/Bin), handheld Android scanner engine, blind 60/40 cycle counts, inter-branch stock transfers (GIT) with driver signatures, and FIFO inventory valuation.

🕒 15 Weeks (3.75 Mo)

MODULE 4

BD 5,113.636

Sales Process, Mobile POS & Branch Cash Control

High-speed counter POS, mobile Android handheld floor sales, dynamic QR cart handoff, credit customer credit-risk limits, verified invoice returns with condition grading, and End-of-Day cash drawer closing locks.

🕒 15 Weeks (3.75 Mo)

MODULE 5**BD 4,431.818****Accounting, General Ledger, AP/AR & VAT Compliance**

5-group Chart of Accounts (COA), real-time double-entry GL, automated AP/AR subledgers, multi-currency treasury, automated Bank Reconciliation (BRS), PDC management, and Bahrain NBR 10% VAT return filings.

🕒 13 Weeks (3.25 Mo)

MODULE 6**BD 4,090.909****Enterprise Administration, Fixed Assets & Fleet**

Centralized branch facility ticketing, fixed asset QR tagging with automated straight-line depreciation, company vehicle fleet logs with Total Cost of Ownership (TCO), and legal document expiry alerts (CR, Leases).

🕒 12 Weeks (3.0 Mo)

MODULE 7**BD 5,454.548****HR Management, Biometrics, Payroll & Gratuity**

Employee 360 master, physical biometric clock-in & geofenced mobile attendance, Bahrain Labour Law leave rules, automated monthly payroll engine, SIO/LMRA/WPS bank export, and End-of-Service Benefit (EOSB) settlements.

🕒 16 Weeks (4.0 Mo)

MODULE 8**BD 1,022.727****Hardware Setup, Scanners, Printers & Cloud Cluster**

3-tier cloud server architecture (Dev / Staging / Prod), RabbitMQ asynchronous background queue, Android PDA scanner listeners, ESC/POS thermal receipt & barcode label printers, and biometric sync daemons.

🕒 3 Weeks (0.75 Mo)

MODULE 9**BD 1,363.636****Executive Management Dashboard & BI Cockpit**

OLAP data warehouse aggregation engine, real-time KPI metrics across all 8 modules, automated anomaly detection, daily 7:00 AM WhatsApp/Email executive digest, and secure C-Suite mobile executive reporting.

🕒 4 Weeks (1.0 Mo)



Complete 35 Milestone Tranches Detailed Breakdown (BHD)

35 MILESTONE
TRANCHES

Specific milestone gates, percentages, exact BHD allocations, and payment triggers for each phase

MOD 1

Module 1: PCode Generation & Item Master Engine (10 Weeks)

BD 3,409.091



MILESTONE 1.1 (25%)

BD 852.273

PCode DB Schema, Data Cleansing Engine & Base Generation Algorithm.

📅 Weeks 1-3

MILESTONE 1.2 (25%)

BD 852.273

Brand / Model / Spec Dynamic Hierarchy & Automated PCode Generator Engine.

📅 Weeks 4-5

MILESTONE 1.3 (25%)

BD 852.273

Universal Cross-Reference / Interchange Matrix & Barcode Tagging Engine.

📅 Weeks 6-8

MILESTONE 1.4 (25%)

BD 852.273

Real-Time Search, Legacy Migration (50,000+ SKUs) & Multi-Branch Final UAT.

📅 Weeks 9-10

MOD 2

Module 2: Vendor & Purchase Management (12 Weeks)

BD 4,090.909



MILESTONE 2.1 (25%)

BD 1,022.727

Supplier Master, Tokenized Vendor Portal & Automated RFQ Engine.

📅 Weeks 1-3

MILESTONE 2.2 (25%)

BD 1,022.727

Multi-Vendor Quotation Evaluation Matrix & Automated PO Split Engine.

📅 Weeks 4-6

MILESTONE 2.3 (25%)

BD 1,022.727

Goods Receipt Note (GRN), Multi-Currency Landed Cost & QC Inspection.

📅 Weeks 7-9

MILESTONE 2.4 (25%)

BD 1,022.727

3-Way Invoice Matching, Vendor SLA Scorecards & Multi-Branch Sign-Off.

📅 Weeks 10-12

MOD 3

Module 3: Store Verification & Stock Control (15 Weeks)

BD 5,113.636



MILESTONE 3.1 (25%)

BD 1,278.409

MILESTONE 3.2 (25%)

BD 1,278.409

Warehouse Spatial Hierarchy (5 Tiers), Bin Matrix & Inward PVN Setup.

📅 Weeks 1-4

Handheld Scanner Engine, Blind 60/40 Cycle Counts & Variance Alerts.

📅 Weeks 5-8

MILESTONE 3.3 (25%)

BD 1,278.409

Inter-Branch Stock Transfers (GIT), Multi-Tier Discrepancy & FIFO Valuation.

📅 Weeks 9-12

MILESTONE 3.4 (25%)

BD 1,278.409

Min/Max Safety Stock Auto-Replenishment, Aging & Final Warehouse Sign-Off.

📅 Weeks 13-15

MOD 4

Module 4: Sales Process, POS & Branch Financial Control (15 Weeks)

BD 5,113.636



MILESTONE 4.1 (25%)

BD 1,278.409

Multi-Branch POS Architecture, Offline Sync & Cash Drawer Session Control.

📅 Weeks 1-4

MILESTONE 4.2 (25%)

BD 1,278.409

Counter & Mobile Handheld POS Billing, Multi-Price Matrix & Barcodes.

📅 Weeks 5-8

MILESTONE 4.3 (25%)

BD 1,278.409

Credit Customer Governance, Credit Limit Locks, Delivery Notes & B2B Invoices.

📅 Weeks 9-12

MILESTONE 4.4 (25%)

BD 1,278.409

Verified Sales Returns, End-of-Day Branch Reconciliation & Audit UAT.

📅 Weeks 13-15

MOD 5

Module 5: Accounting & Financial Management (13 Weeks)

BD 4,431.818



MILESTONE 5.1 (25%)

BD 1,107.955

Dynamic Chart of Accounts (COA), Cost Centers & Auto Journal Engine.

📅 Weeks 1-3

MILESTONE 5.2 (25%)

BD 1,107.955

AP / AR Subledgers, Aging Analysis, Statements & Credit Lock Rules.

📅 Weeks 4-6

MILESTONE 5.3 (25%)

BD 1,107.955

Multi-Currency Treasury, Automated Bank Reconciliation & Cashflow Engine.

📅 Weeks 7-10

MILESTONE 5.4 (25%)

BD 1,107.955

Bahrain NBR 10% VAT Return Generator, Balance Sheet, P&L & Final Sign-Off.

📅 Weeks 11-13

MOD 6

Module 6: Enterprise Administration & Fixed Assets (12 Weeks)

BD 4,090.909



MILESTONE 6.1 (25%)**BD 1,022.727**

Facility Management, Maintenance Ticketing, Utility Logs & Lease Alerts.

 Weeks 1-3**MILESTONE 6.2 (25%)****BD 1,022.727**

Fixed Asset Master, QR Tagging, Custodian Tracking & Depreciation Engine.

 Weeks 4-6**MILESTONE 6.3 (25%)****BD 1,022.727**

Fleet Logistics, Vehicle Bookings, Fuel Reconciliation & TCO Analytics.

 Weeks 7-9**MILESTONE 6.4 (25%)****BD 1,022.727**

Staff Locker Control, Secure Digital Document Vault & CR Renewal Alerts.

 Weeks 10-12**MOD 7****Module 7: Human Resource Management & Payroll (16 Weeks)****BD 5,454.548****MILESTONE 7.1 (25%)****BD 1,363.637**

Org Structure, Employee Master, Document Vault & Digital Onboarding.

 Weeks 1-4**MILESTONE 7.2 (25%)****BD 1,363.637**

Biometric Attendance, Shift Rosters & Bahrain Labour Law Leave System.

 Weeks 5-8**MILESTONE 7.3 (25%)****BD 1,363.637**

Automated Payroll, SIO/LMRA Compliance, WPS/CBB Export & Loan Advances.

 Weeks 9-12**MILESTONE 7.4 (25%)****BD 1,363.637**

Performance KPIs, ESS Self-Service Portal, EOSB Gratuity & Production Go-Live.

 Weeks 13-16**MOD 8****Module 8: Hardware Integration & Cloud Cluster (3 Weeks)****BD 1,022.727****MILESTONE 8.1 (33.33%)****BD 340.909**

3-Tier Server Environment (Dev/Stage/Prod), RabbitMQ & Backup Automation.

 Week 1**MILESTONE 8.2 (33.33%)****BD 340.909**

Handheld Android QR Scanners, ESC/POS Printers & Biometric Clocks.

 Week 2**MILESTONE 8.3 (33.34%)****BD 340.909**

Offline Auto-Sync Engine, Multi-Branch Stress Testing & Go-Live.

 Week 3

MILESTONE 9.1 (25%)

BD 340.909

Data Warehouse OLAP Schema, Non-Blocking ETL Pipelines & KPI Models.

 Week 1**MILESTONE 9.2 (25%)**

BD 340.909

Commercial, Stock Velocity, Warehouse Verification & Vendor SLA Cockpit.

 Week 2**MILESTONE 9.3 (25%)**

BD 340.909

Financial Liquidity, VAT, HR Payroll, Fixed Assets & Infra Health Analytics.

 Week 3**MILESTONE 9.4 (25%)**

BD 340.909

Management Filtering, 7:00 AM WhatsApp/Email Digest & Final Sign-Off.

 Week 4



Dedicated Engineering Team & Transparent Rate Matrix

TEAM MONTHLY RATE: BD
1,363.636

Full-time dedicated 5-person engineering team rate card in Bahraini Dinars (BHD)

All modules are estimated strictly using the agreed dedicated team rate matrix in Bahraini Dinars (BHD). Total dedicated team capacity: 160 hours / month per engineer.

**Project Manager &
Enterprise Solutions
Architect**

BD 545.455

BD 136.364 / Wk • BD 3.409 / Hr

100% Dedicated

**Back-End Lead Engineer
(PHP MVC / REST APIs)**

BD 227.273

BD 56.818 / Wk • BD 1.420 / Hr

100% Dedicated

**Front-End Lead Engineer
(React / UI Design Tokens)**

BD 227.273

BD 56.818 / Wk • BD 1.420 / Hr

100% Dedicated

**Database & Cloud DevOps
Specialist**

BD 204.545

BD 51.136 / Wk • BD 1.278 / Hr

100% Dedicated

**QA Automation & Multi-
Branch Test Lead**

BD 159.091

BD 39.773 / Wk • BD 0.994 / Hr

100% Dedicated



Total Project Investment & Milestone Schedule in Bahraini Dinars (BHD)

GRAND TOTAL: BD 34,090.910

Consolidated commercial breakdown and 4-milestone payment tranche distribution for Popular Auto Spare Co.

MODULE REFERENCE	MODULE DOMAIN	TOTAL (BHD)	TRANCHE 1 (BHD)	TRANCHE 2 (BHD)	TRANCHE 3 (BHD)	TRANCHE 4 (BHD)
MS-001	PCode Generation Module	BD 3,409.091	BD 852.273	BD 852.273	BD 852.273	BD 852.273
MS-002	Vendor & Purchase Module	BD 4,090.909	BD 1,022.727	BD 1,022.727	BD 1,022.727	BD 1,022.727
MS-003	Store Verification Module	BD 5,113.636	BD 1,278.409	BD 1,278.409	BD 1,278.409	BD 1,278.409
MS-004	Sales Process & POS Module	BD 5,113.636	BD 1,278.409	BD 1,278.409	BD 1,278.409	BD 1,278.409
MS-005	Accounts & VAT Module	BD 4,431.818	BD 1,107.955	BD 1,107.955	BD 1,107.955	BD 1,107.955
MS-006	Enterprise Admin & Fleet	BD 4,090.909	BD 1,022.727	BD 1,022.727	BD 1,022.727	BD 1,022.727
GRAND TOTAL INVESTMENT (BHD)		BD 34,090.910	BD 8,707.955	BD 8,707.955	BD 8,707.955	BD 7,967.04

MODULE REFERENCE	MODULE DOMAIN	TOTAL (BHD)	TRANCHE 1 (BHD)	TRANCHE 2 (BHD)	TRANCHE 3 (BHD)	TRANCHE 4 (BHD)
MS-007	HRM & Payroll Module	BD 5,454.548	BD 1,363.637	BD 1,363.637	BD 1,363.637	BD 1,363.637
MS-008	Hardware & Cloud Setup	BD 1,022.727	BD 340.909	BD 340.909	BD 340.909	—
MS-009	Executive BI Dashboard	BD 1,363.636	BD 340.909	BD 340.909	BD 340.909	BD 340.909
GRAND TOTAL INVESTMENT (BHD)		BD 34,090.910	BD 8,707.955	BD 8,707.955	BD 8,707.955	BD 7,967.04

Governance, SLA & Payment Terms

ENTERPRISE SLA

Contractual milestones, UAT acceptance gates, and IP protection protocols

1. Milestone-Driven Invoicing & Payment Tranches

- **Zero Blind Upfront Risk:** Invoices are raised strictly upon completion, demonstration, and staging sign-off of each milestone tranche (25% or 33.3% per gate).
- **15-Day Verification & UAT Grace Period:** The client is entitled to a 15-calendar-day UAT review window per milestone to test all specified acceptance criteria before signing off.
- **Change Request (CR) Protocol:** Any features outside the defined scope are logged via SaNDS Lab Portal and quoted using the transparent hourly rate card (BD 8.523 / Hour).

2. Source Code Ownership, Security & Warranty

- **100% Intellectual Property (IP) Ownership:** Upon settlement of contract invoices, full proprietary source code, DB schemas, and deployment scripts belong exclusively to Popular Auto Spare Co.
- **Zero Third-Party Recurring Royalties:** Built on open-standard enterprise stacks (PHP 8.3 MVC, MySQL InnoDB, React 18, RabbitMQ, SQLite) with zero vendor lock-in.
- **12-Month Post-Launch Bug Warranty:** 365 days of comprehensive bug-fixing SLA and security patch support following the final production deployment of each module.

FINAL STAGE

OFFICIAL ACCEPTANCE & DIGITAL VERIFICATION

Stakeholder Authorization & Digital Sign-Off Console

By affixing digital signatures and authorizations below, all participating executive and technical stakeholders ratify the deliverables, implementation timeline, SLA commitments, and financial disbursement gates for **Master 9-Module ERP Milestone & Budget Roadmap (SL-POP-ERP-SUMMARY-001)**:

1. Engineering Preparation & Stakeholder Verification Matrix

DEVELOPER / ARCHITECT

Ancy Varghese Thekkan

Software Architect, SaNDS Lab

✓ PREPARED (25-SEPTEMBER-2026)

COMPLETED

VERIFIED BY (DEVELOPER)

Ajit Kumar KV

CEO & Managing Director, SaNDS Lab

✓ VERIFIED (30-SEPTEMBER-2026)

CHECKED & VERIFIED

CONSULTANT REVIEW

UniGlobal Business Solutions

Project Management Consultant

🕒 In Active Review

IN REVIEW

CLIENT APPROVAL

Popular Auto Spare & A/C Parts

Authorized Executive Signatory

🕒 Pending Approval

PENDING SIGNATURE

2. Executive Multi-Party Digital Acceptance Console

Popular Auto Spare Co. W.L.L

Client Executive Sign-off & Commercial Approval

✍ No Signature on Record

SaNDS Lab Middle East W.L.L

Lead Solutions Architect & Engineering Director

✍ No Signature on Record

PENDING SIGN-OFF

PENDING SIGN-OFF

UniGlobal Consultancy

Executive Strategic Advisor & Governance Lead

 No Signature on Record

PENDING SIGN-OFF